

Question:

A 38-year-old man comes to the physician because of progressive pain and swelling of his left knee for the past 2 days. He has been taking ibuprofen for the past 2 days without improvement. Four days ago, he scraped his left knee while playing baseball. He has a 2-month history of progressive pain and stiffness in his back. The pain starts after waking up and lasts for 20 minutes. He has type 2 diabetes mellitus. His older sister has rheumatoid arthritis. He is 170 cm (5 ft 7 in) tall and weighs 91 kg (201 lb); BMI is 31.5 kg/m². Temperature is 39°C (102.2°F), pulse is 90/min, and blood pressure is 135/85 mm Hg. Examination shows an erythematous, tender, and swollen left knee; range of motion is limited. There are abrasions over the lateral aspect of the left knee. The remainder of the examination shows no abnormalities. Laboratory studies show a leukocyte count of 13,500/mm³ and an erythrocyte sedimentation rate of 70 mm/h. Which of the following is the most appropriate next step in management?

Answer Choices:

- A. Bone scan
- B. MRI of the left knee
- C. Blood cultures
- D. Arthrocentesis of the left knee

Reasoning + Answer:

<think>

The patient presents with symptoms suggestive of a septic joint , given the acute onset of swelling and severe pain after an injury , and signs of systemic inflammation with fever and elevated ESR . Arthrocentesis is the first step in diagnosing a septic joint by identifying infection .

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<answer>D . Arthrocentesis of the left knee</answer>

